CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CE) Examination

November 2013

CE305 Computer Network & Internetworking Layers

Date: 03.12.2013, Tuesday | **Time: 01:30 p.m. To 04:30 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Answer the following questions. **[06]**

1. With proper example explain Stop N Wait protocol.
2. Explain Optimality Principle and give example.

(b) Explain ATM reference model with diagram. **[05]**

Q - 2 (a) Explain Link State Routing Protocol in detail. **[04]**

(b) Explain Go Back N ARQ. **[04]**

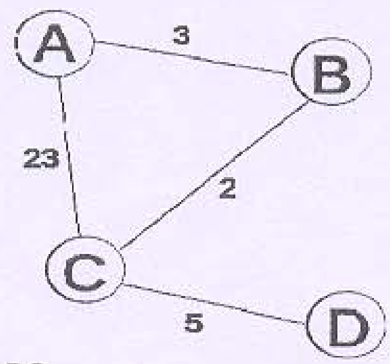
(c) Calculate VRC and LRC for the following bit pattern using even parity for 1: **[04]**
←0011101 1100111 1111111 0000000
Also compare performance of both methods.

OR

Q - 2 (a) Differentiate following. **[04]**

1. Connection-Oriented and Connectionless services.
2. IPV4 and IPV6.

(b) For given topology generate Routing tables for any 3 nodes using Distance Vector Routing algorithm. **[04]**



(c) Explain Selective Repeat ARQ. **[04]**

Q - 3 (a) Explain Bit Stuffing method for framing. **[04]**

(b) Differentiate: Virtual Circuit and Datagram Subnets. **[04]**

(c) Draw the figure for IPV4 Header format and explain each term in brief. **[04]**

OR

(c) Explain HDLC frame format. Also explain control fields in detail. **[04]**

SECTION – II

Q - 4 (a) Represent 1 0 1 0 0 1 1 0 1 0 1 0 0 1 1 1 in Manchester and Differential Manchester encoding. [04]

(b) **Answer the following questions.** [07]

1. Explain Multiplexing in transport layer. [03]
2. Explain RTS and CTS with example. [04]

Q - 5 (a) Which devices are used for networking? Explain any four devices in brief. [05]

(b) **Do as Directed.** [07]

1. Draw the Frame format for Ethernet and explain each term in detail. [04]
2. Explain ALOHA and Slotted ALOHA. [03]

OR

Q - 5 (a) Explain TCP Connection Management in normal case and when collision occurs. [06]

(b) Explain transport layer Quality of Service parameters. [06]

Q - 6 (a) Explain TCP header in detail. [06]

(b) What is DHCP? Why is it used? [06]

OR

Q - 6 (a) Explain: Domain Name System (DNS). [06]

(b) Write down short note on E-mail. [06]

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